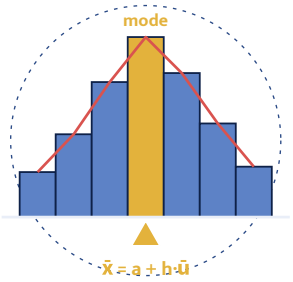
MT Classes, Motihari

Statistics

Class 10 Maths, Chapter 13 (NCERT, CBSE 2026-27)



Mean, mode and median of grouped data, made simple and made fun: every table of the textbook worked out, pictures that show *why* each formula works, six calculators, an **arcade of 8 games and puzzles** (see-saw balance, memory match, speed round, frequency detective and more), real CBSE board questions 2023–2026, every NCERT question solved and a 20-question quiz.

📌 Chapter snapshot

- Chapter 13 of NCERT Class 10 Mathematics (reprint 2026-27): sections 13.1 to 13.5, 8 solved examples and **three exercises (9 + 6 + 7 = 22 questions)**.
- Part of the unit **Statistics and Probability** (11 marks in the 80-mark paper). Statistics alone usually gives **6 to 9 marks**: one or two 1-mark MCQs (often the **empirical relation** or the **modal / median class**), and either a **5-mark** question (missing frequency, mean and mode) or a **4-mark case study** (median, mode, mean).
- Asked every year: **3 Median = Mode + 2 Mean** (2025, 2026), **missing frequencies from the mean** (2025, 2026), **median and mode in a case study** (2023, 2024), **modal class from a "less than" table** (2023).
- Ogives (cumulative frequency graphs) are no longer in the chapter. The cumulative frequency **table** is, and you need it for the median.

13.1 From Class IX to grouped data

In Class IX you found the **mean, median and mode** of small lists of numbers. Real data is big: the marks of 2 lakh students, the heights of every girl in a district, the electricity bills of a whole town. So we **group** it into classes, and this chapter teaches how to find the three "centres" of such **grouped data**.

$\bar{x}$ Mean the average

Sum of all values ÷ number of values. Uses every observation.

Median the middle one

The value that splits the data, arranged in order, into two equal halves.

Mode the most popular

The value that occurs most often (highest frequency).

These three are called **measures of central tendency**: each tries to give **one number** that represents the whole data.

Words you must know

Word	Meaning	Example (class 40–55)
Class interval	A range of values, written lower limit – upper limit	40–55
Lower / upper class limit	The smaller / larger end of the class	40 and 55
Class size (width) h	Upper limit − lower limit	55 − 40 = 15
Class mark (mid-point) x_i	$\frac{\text{upper limit + lower limit}}{2}$	$\frac{40+55}{2} = 47.5$
Frequency f_i	How many observations fall in the class	7 students
Cumulative frequency cf	Frequency of the class + frequencies of all classes before it	2 + 3 + 7 = 12

🤔 **Where does a value on the boundary go?**

In NCERT, a value equal to an upper limit goes into the **next** class. So 40 marks is counted in **40–55**, not in 25–40. This is the meaning of "10–25, 25–40, ..." (classes that share an end point).

13.2 Mean of grouped data: the direct method

If $x_1, x_2, ..., x_n$ occur with frequencies $f_1, f_2, ..., f_n$, then x_1 occurs f_1 times, x_2 occurs f_2 times, and so on.

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + ... + f_nx_n}{f_1 + f_2 + ... + f_n} = \frac{\Sigma f_ix_i}{\Sigma f_i}$$

Σ (capital sigma, a Greek letter) means "add them all up", for i = 1 to n.

Example 1: marks of 30 students (ungrouped)

The marks (out of 100) of 30 students of Class X:

Marks (x_i)	10	20	36	40	50	56	60	70	72	80	88	92	95
Students (f_i)	1	1	3	4	3	2	4	4	1	1	2	3	1

Sol. Multiply each x_i by its f_i and add.

Table 13.1

Marks obtained (x_i)	Number of students (f_i)	$f_i x_i$
10	1	10
20	1	20
36	3	108
40	4	160
50	3	150
56	2	112
60	4	240
70	4	280
72	1	72
80	1	80
88	2	176
92	3	276
95	1	95
Total	**$\Sigma f_i = 30$**	**$\Sigma f_i x_i = 1779$**

$$\bar{x} = \frac{1779}{30} = 59.3$$

The mean mark is **59.3**.

Now group the same data

Make classes of width 15: 10–25, 25–40, ..., 85–100. Remember: the 4 students with exactly 40 marks go into **40–55**.

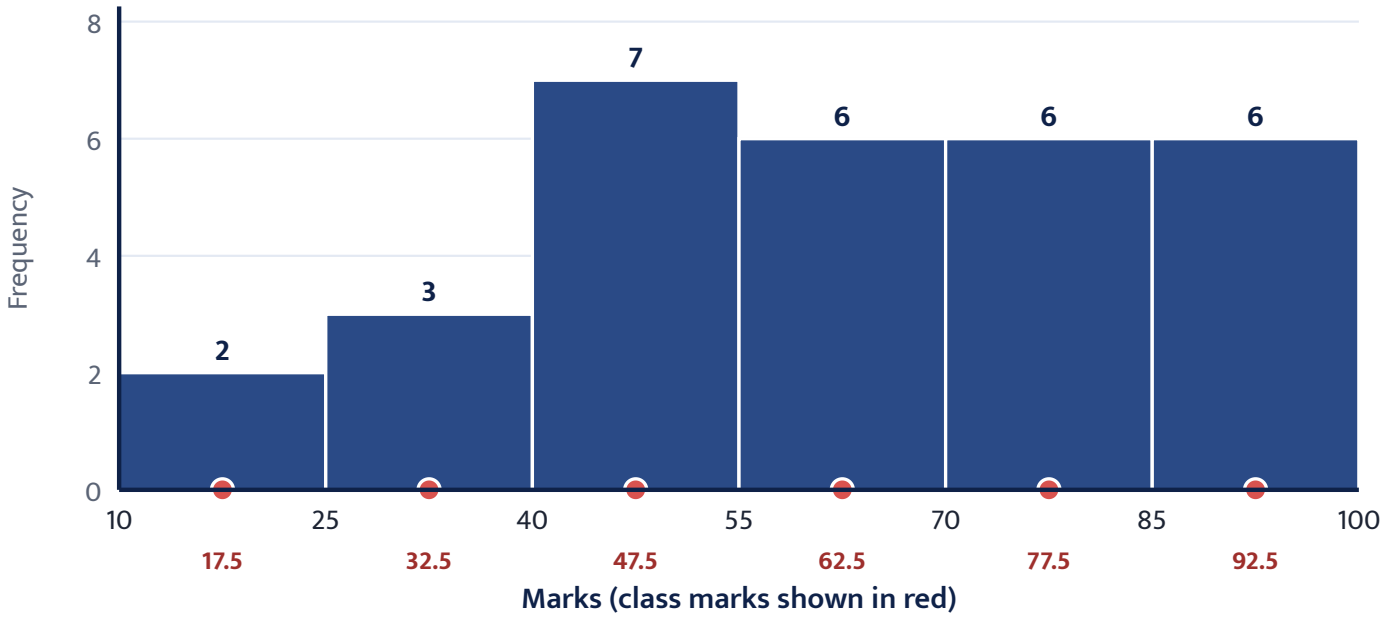
Table 13.2

Class interval	10–25	25–40	40–55	55–70	70–85	85–100
Number of students	2	3	7	6	6	6

Inside a class we no longer know the exact marks. So we **assume** that every value in a class sits at its centre, the **class mark**:

$$\textbf{Class mark } x_i = \frac{\textbf{Upper class limit + Lower class limit}}{\textbf{2}}$$

For 10–25: $\frac{10 + 25}{2} = 17.5$. The class marks become our x_i.



Picture of Table 13.2. Each bar is a class. The red dot under each bar is its class mark: we pretend all the students of that bar scored this value.

Table 13.3

Class interval	Number of students (f_i)	Class mark (x_i)	$f_i x_i$
10–25	2	17.5	35
25–40	3	32.5	97.5
40–55	7	47.5	332.5
55–70	6	62.5	375
70–85	6	77.5	465

Class interval	Number of students (f_i)	Class mark (x_i)	$f_i x_i$
85–100	6	92.5	555
Total	**$\Sigma f_i = 30$**		**$\Sigma f_i x_i = 1860.0$**

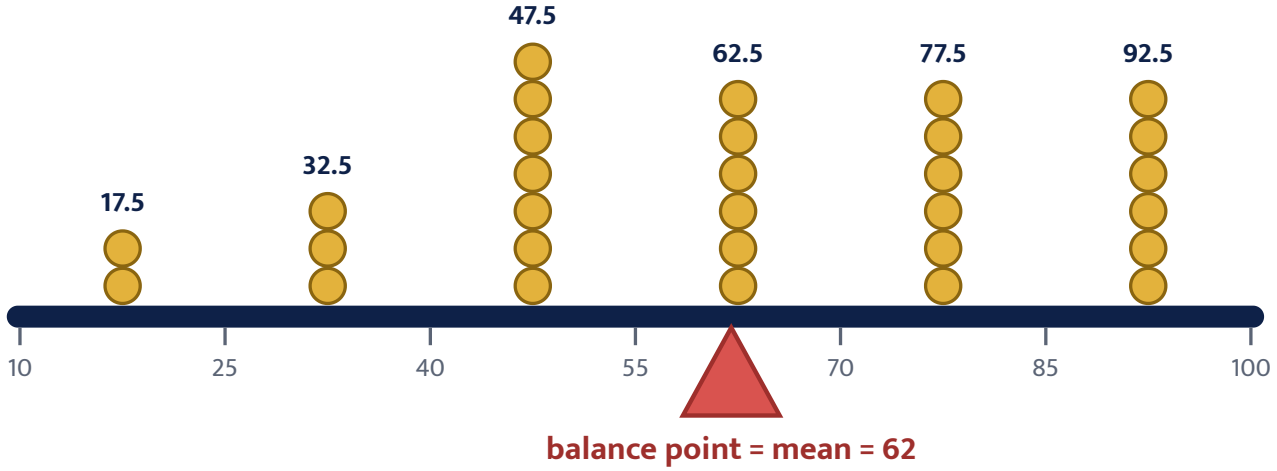
$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1860.0}{30} = 62$$

This is called the **direct method**.

🤔 Why 59.3 before and 62 now?

Same students, same formula, different answers! 59.3 is the **exact** mean (Table 13.1). 62 is an **approximate** mean, because of the **mid-point assumption** in Table 13.3. Grouping always loses a little information.

Each coin is one student, placed at the class mark of their class.



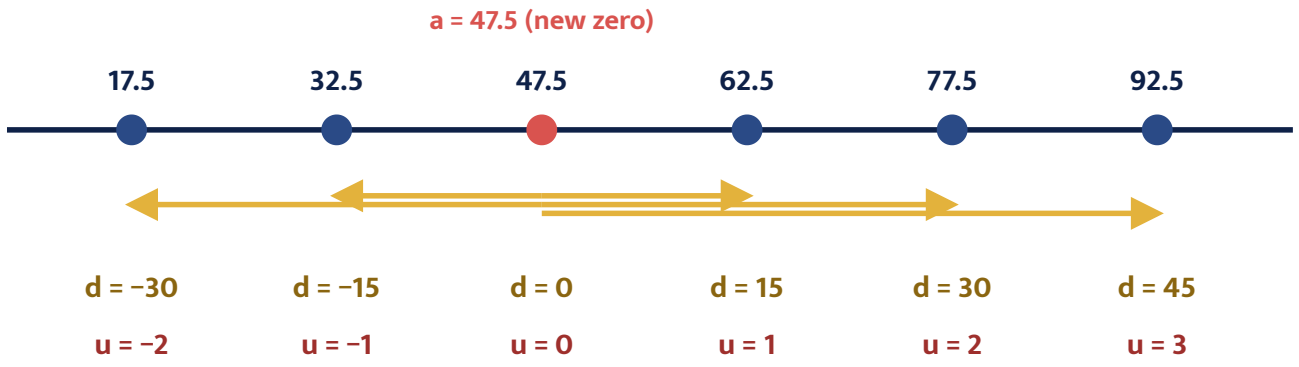
Why it is called the "average". Put one coin for every student at the class mark of their class on a weightless beam. The beam balances exactly at the mean, 62. More coins on one side pull the balance point that way. (You will play this as a puzzle in the Stats Arcade.)

13.2 Assumed mean and step-deviation methods

When x_i and f_i are big numbers, multiplying them is slow and error-prone. We cannot change the f_i, but we **can make the x_i smaller** by subtracting a fixed number from each of them.

Assumed mean method

1. Choose one of the x_i as the **assumed mean** a. Pick one near the **centre** to keep numbers small. Here a = 47.5 (62.5 also works).
2. Find the **deviations** $d_i = x_i - a$.
3. Find $f_i d_i$ and add them.
4. Mean of the deviations $\bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i}$, and $\bar{x} = a + \bar{d}$.



d = distance from a · u = d ÷ h (h = 15)

Shifting the zero. Taking a = 47.5 moves the zero of the number line to 47.5. The deviations d are now small numbers, and all of them are multiples of 15, the class size, so dividing by 15 gives the tiny numbers u.

Table 13.4

Class interval	f_i	x_i	$d_i = x_i - 47.5$	$f_i d_i$
10–25	2	17.5	−30	−60
25–40	3	32.5	−15	−45
40–55	7	47.5	0	0
55–70	6	62.5	15	90
70–85	6	77.5	30	180
85–100	6	92.5	45	270
Total	**$\Sigma f_i = 30$**			**$\Sigma f_i d_i = 435$**

Why $\bar{x} = a + \bar{d}$ (the proof in the book)

$$\bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i} = \frac{\Sigma f_i (x_i - a)}{\Sigma f_i} = \frac{\Sigma f_i x_i}{\Sigma f_i} - \frac{\Sigma f_i a}{\Sigma f_i} = \bar{x} - a \times \frac{\Sigma f_i}{\Sigma f_i} = \bar{x} - a$$

So $\bar{x} = a + \bar{d}$, that is

$$\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$$

$$\bar{x} = 47.5 + \frac{435}{30} = 47.5 + 14.5 = 62$$

⭐ Activity 1: does the choice of a matter?

Take a = 17.5, then 32.5, then 62.5 … You get **62 every time**. The value of the mean **does not depend on the choice of a**, because whatever we subtract we add back. Try it in Tool 2 below.

Step-deviation method

In Table 13.4 every d_i is a multiple of 15 (the class size). Divide them by 15 to get even smaller numbers:

$$u_i = \frac{x_i - a}{h}$$

a = assumed mean, h = class size

Table 13.5

Class interval	f_i	x_i	$d_i = x_i - a$	$u_i = \frac{x_i - a}{h}$	$f_i u_i$
10–25	2	17.5	−30	−2	−4
25–40	3	32.5	−15	−1	−3
40–55	7	47.5	0	0	0
55–70	6	62.5	15	1	6
70–85	6	77.5	30	2	12
85–100	6	92.5	45	3	18
Total	**$\Sigma f_i = 30$**				**$\Sigma f_i u_i = 29$**

Why $\bar{x} = a + h\,\bar{u}$

$$\bar{u} = \frac{\Sigma f_i u_i}{\Sigma f_i} = \frac{\Sigma f_i (x_i - a)/h}{\Sigma f_i} = \frac{1}{h}\left(\frac{\Sigma f_i x_i}{\Sigma f_i} - a\,\frac{\Sigma f_i}{\Sigma f_i}\right) = \frac{1}{h}(\bar{X} - a)$$

So $h\bar{u} = \bar{X} - a$, that is

$$\bar{x} = a + h \times \left(\frac{\sum f_i u_i}{\sum f_i}\right)$$

$$\bar{x} = 47.5 + 15 \times \frac{29}{30} = 47.5 + 14.5 = 62$$

✍️ **Remember**

- Step-deviation is handy when **all the d_i have a common factor**.
- All three methods give the **same mean**. The assumed mean and step-deviation methods are just **simplified forms of the direct method**.
- $\bar{x} = a + h\bar{u}$ still holds if a and h are **any non-zero numbers** with $u_i = \frac{x_i - a}{h}$. That is why it works even for unequal classes (Example 3).

Ex.

Examples 2 and 3: choosing a method

Example 2: female teachers in rural primary schools

Percentage of female teachers in the rural primary schools of states and union territories of India (Seventh All India School Education Survey, NCERT). Find the mean by all three methods.

Percentage	15–25	25–35	35–45	45–55	55–65	65–75	75–85
States / U.T.	6	11	7	4	4	2	1

Sol. Class marks 20, 30, ..., 80. Take a = 50, h = 10.

Table 13.7

% female teachers	f_i	x_i	$d_i = x_i - 50$	$u_i = d_i/10$	f_ix_i	f_id_i	f_iu_i
15–25	6	20	−30	−3	120	−180	−18
25–35	11	30	−20	−2	330	−220	−22
35–45	7	40	−10	−1	280	−70	−7
45–55	4	50	0	0	200	0	0
55–65	4	60	10	1	240	40	4
65–75	2	70	20	2	140	40	4
75–85	1	80	30	3	80	30	3
Total	**35**				**1390**	**−360**	**−36**

Direct

$$\bar{x} = \frac{1390}{35} = 39.71$$

Assumed mean

$$\bar{x} = 50 + \frac{-360}{35} = 39.71$$

Step-deviation

$$\bar{x} = 50 + 10 \times \frac{-36}{35} = 39.71$$

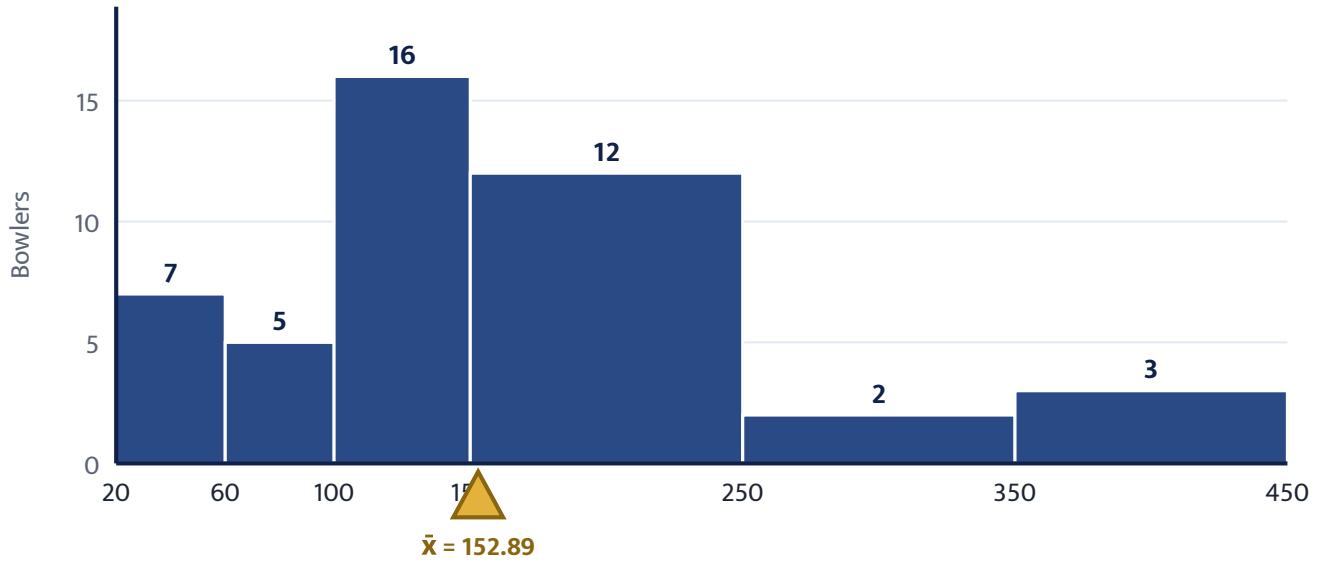
The mean percentage of female teachers is **39.71**.

⭐ Which method should I use?

- x_i and f_i **small** → **direct method**.
- x_i and f_i **large** → **assumed mean** or **step-deviation**.
- Class sizes **unequal** and x_i large → still step-deviation, taking h as a **suitable common divisor of all the d_i**.

Example 3: wickets taken by bowlers (unequal classes)

Wickets	20–60	60–100	100–150	150–250	250–350	350–450
Bowlers	7	5	16	12	2	3



Number of wickets (classes of different widths)

Unequal classes. The classes are 40, 40, 50, 100, 100 and 100 wide, so h is not one fixed class size. We still use step-deviation with a = 200 and h = 20, because 20 divides every d_i except −75 (u = −3.75 is fine).

Sol. a = 200, h = 20.

Table 13.8

Wickets	Bowlers f_i	x_i	$d_i = x_i - 200$	$u_i = d_i/20$	f_iu_i
20–60	7	40	−160	−8	−56
60–100	5	80	−120	−6	−30
100–150	16	125	−75	−3.75	−60
150–250	12	200	0	0	0
250–350	2	300	100	5	10
350–450	3	400	200	10	30
Total	**45**				**−106**

$$\bar{x} = 200 + 20 \times \frac{-106}{45} = 200 - 47.11 = 152.89$$

On average, these 45 bowlers took **152.89 wickets** in one-day cricket.

🧪 Activity 2 (do it in class)

Three groups collect (1) everyone's maths marks in the last exam, (2) the daily maximum temperature in Motihari for 30 days, (3) everyone's height in cm. Make a grouped frequency table and find the mean by the method you like best. Then try Tool 1 with your own numbers.

13.3 Mode of grouped data

The **mode** is the value with the **maximum frequency**. If two or more values share the maximum, the data is **multimodal** (we only take problems with one mode).

Example 4: ungrouped data first

Wickets taken by a bowler in 10 matches: 2, 6, 4, 5, 0, 2, 1, 3, 2, 3.

Wickets	0	1	2	3	4	5	6
Matches	1	1	**3**	2	1	1	1

2 wickets came in the most matches (3), so the **mode is 2**.

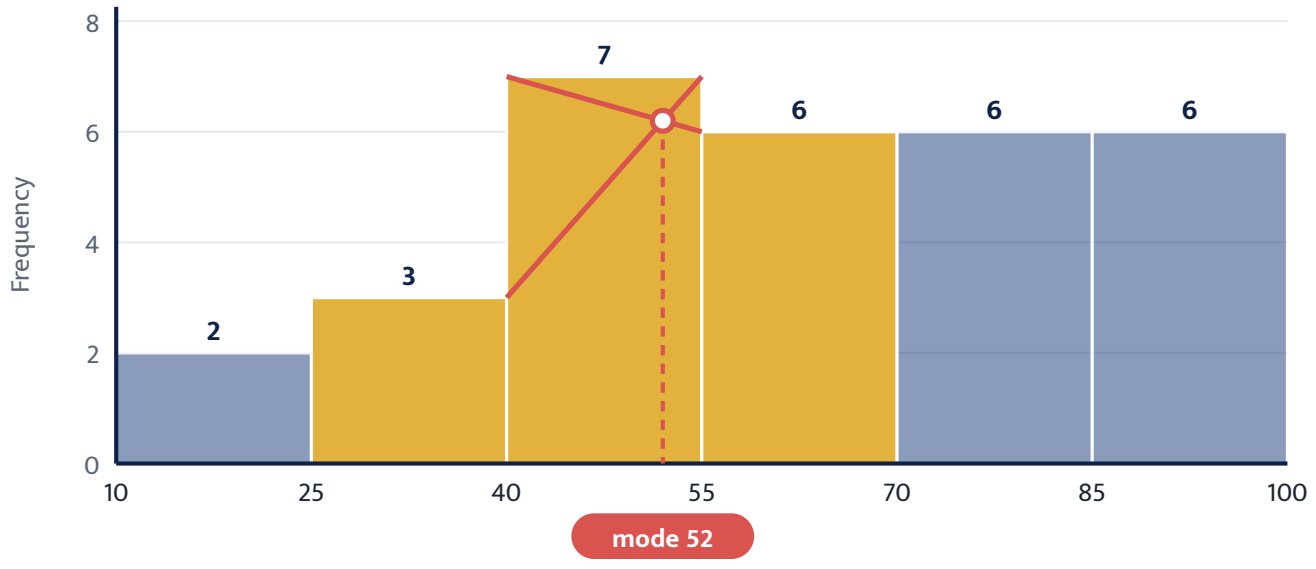
The modal class and the mode formula

In grouped data we can only spot the class with the highest frequency, the **modal class**. The mode is a value **inside** it:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Only for classes of equal size.

- l lower limit of the modal class
- h size of the class interval
- f_1 frequency of the modal class
- f_0 frequency of the class **before** (preceding) the modal class
- f_2 frequency of the class **after** (succeeding) the modal class



Marks (Table 13.3)

Where the formula comes from. In the tallest bar (40–55), join its top-left corner to the top-left corner of the next bar, and its top-right corner to the top-right corner of the bar before. The lines cross above the mode. If f_0 is smaller than f_2, the crossing leans to the right. Working this out with similar triangles gives exactly the formula above: here $40 + \frac{7-3}{14-3-6} \times 15 = \mathbf{52}$.

Example 5: family size of 20 households

Family size	1–3	3–5	5–7	7–9	9–11
Families	7	8	2	2	1

– **Sol.** **Modal class 3–5 (frequency 8).**

$l = 3, h = 2, f_1 = 8, f_0 = 7, f_2 = 2.$

$$\text{Mode} = 3 + \left(\frac{8-7}{16-7-2}\right) \times 2 = 3 + \frac{2}{7} = 3.286$$

Example 6: mode of Table 13.3, compared with the mean

– **Sol.** **Modal class 40–55 (7 students).**

$l = 40, h = 15, f_1 = 7, f_0 = 3, f_2 = 6.$

$$\text{Mode} = 40 + \left(\frac{7-3}{14-6-3}\right) \times 15 = 40 + 12 = 52$$

The **largest number of students scored about 52**, while **on average a student scored 62** (the mean).

🤔 Remarks from the book

- Here mode < mean, but in other data the mode can be equal to or bigger than the mean.
- "Average marks of all students" → use the **mean**. "The marks most students got" → use the **mode**.
- The mode can be found for unequal classes too, but not in Class 10.

13.4 Median of grouped data

The **median** is the **middle-most** observation. For ungrouped data in ascending order:

n odd

Median = $(\frac{n+1}{2})^{th}$ observation

n even

Median = average of the $(\frac{n}{2})^{th}$ and $(\frac{n}{2}+1)^{th}$

Ungrouped example: marks of 100 students (out of 50)

Marks	20	29	28	33	42	38	43	25
Students	6	28	24	15	2	4	1	20

Arrange in ascending order and keep adding the frequencies. This running total is the **cumulative frequency**.

Table 13.11

Marks obtained	Number of students	Cumulative frequency
20	6	6
25	20	26
28	24	**50**
29	28	**78**
33	15	93
38	4	97
42	2	99
43	1	100
Total	**100**	

n = 100 (even), so we need the 50th and 51st observations. The cf reaches 50 at 28, so the **50th is 28**; the 51st to 78th are all 29.

Median = $\frac{28+29}{2}$ = 28.5

About 50% of the students scored less than 28.5 and 50% scored more. Columns 1 and 3 together are called a cumulative frequency table.

Grouped data: two kinds of cumulative frequency

Marks (out of 100) of 53 students:

Table 13.12

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Students	5	3	4	3	3	4	7	9	7	8

Table 13.13: "less than" uses the upper limits

Marks	cf (less than type)
Less than 10	5
Less than 20	5 + 3 = 8
Less than 30	8 + 4 = 12
Less than 40	12 + 3 = 15
Less than 50	15 + 3 = 18
Less than 60	18 + 4 = 22
Less than 70	22 + 7 = 29
Less than 80	29 + 9 = 38
Less than 90	38 + 7 = 45
Less than 100	45 + 8 = 53

Table 13.14: "more than" uses the lower limits

Marks	cf (more than or equal to)
≥ 0	53
≥ 10	53 − 5 = 48
≥ 20	48 − 3 = 45
≥ 30	45 − 4 = 41
≥ 40	41 − 3 = 38
≥ 50	38 − 3 = 35
≥ 60	35 − 4 = 31

Marks	cf (more than or equal to)
≥ 70	31 − 7 = 24
≥ 80	24 − 9 = 15
≥ 90	15 − 7 = 8

Joining Table 13.12 and 13.13 gives the table we use for the median:

Table 13.15

Marks	Number of students (f)	Cumulative frequency (cf)
0–10	5	5
10–20	3	8
20–30	4	12
30–40	3	15
40–50	3	18
50–60	4	22
60–70	7	**29**
70–80	9	38
80–90	7	45
90–100	8	53

Median class and the median formula

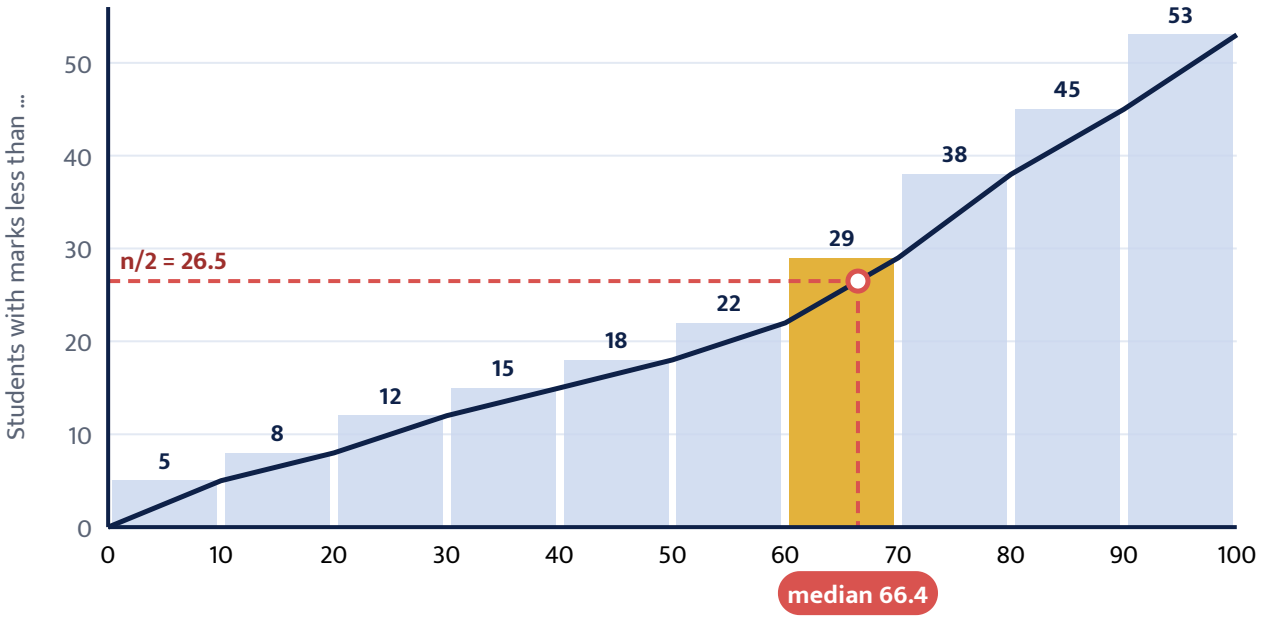
1. Find $n = \Sigma f$ and $\frac{n}{2}$. Here $n = 53$, $\frac{n}{2} = 26.5$.
2. The **median class** is the class whose cf is **greater than and nearest to** $\frac{n}{2}$: cf 29 > 26.5, so the median class is **60–70**.
3. Read l, cf, f, h and put them in the formula.

$$\textbf{Median} = l + \left(\frac{n/2 - cf}{f}\right) \times h$$

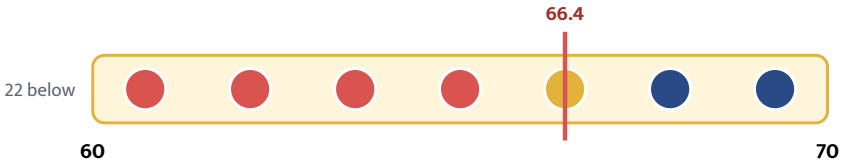
l = lower limit of the median class · n = number of observations · cf = cumulative frequency of the class **preceding** the median class · f = frequency of the median class · h = class size

$$\text{Median} = 60 + \left(\frac{26.5 - 22}{7}\right) \times 10 = 60 + \frac{45}{7} = 66.4$$

About half the students scored less than 66.4 and half scored more.



Seeing the median. The bars show "less than" cumulative frequencies. Go across from 26.5 (half of 53): you hit the graph inside the gold class 60–70. Coming down gives the median, 66.4. (This curve is called an **ogive**; drawing it is no longer in the syllabus, but it shows why the formula works.)



The 7 students of 60–70 are spread evenly. We need 26.5 − 22 = 4.5 of them, so go 4.5/7 of the width: 60 + (4.5 ÷ 7) × 10 = 66.4

Zooming into the median class. The formula assumes the f students of the median class are spread evenly across it, and walks in just far enough to reach the $\frac{n}{2}^{th}$ student.

Ex.

Examples 7 and 8

Example 7: From a "less than" table to the median

Heights of 51 girls of Class X: less than 140 → 4, less than 145 → 11, less than 150 → 29, less than 155 → 40, less than 160 → 46, less than 165 → 51. Find the median height.

Sol. First turn the cumulative frequencies back into class frequencies by **subtracting.**

Table 13.16

Class interval	Frequency	Cumulative frequency
Below 140	4	4
140–145	11 − 4 = 7	11
145–150	29 − 11 = **18**	**29**
150–155	40 − 29 = 11	40
155–160	46 − 40 = 6	46
160–165	51 − 46 = 5	51

$n = 51$, $\frac{n}{2} = 25.5$ lies in **145–150**. $l = 145$, $cf = 11$, $f = 18$, $h = 5$.

$$\textbf{Median} = 145 + \left(\frac{25.5 - 11}{18}\right) \times 5 = 145 + \frac{72.5}{18} = 149.03 \text{ cm}$$

About 50% of the girls are shorter than 149.03 cm and 50% are taller.

Example 8: Find the missing frequencies x and y

The median of the data is 525 and the total frequency is 100.

Class	0–100	100–200	200–300	300–400	400–500	500–600	600–700	700–800	800–900
Frequency	2	5	x	12	17	20	y	9	7

Sol. **Write the cf column in terms of x and y.**

Class	Frequency	Cumulative frequency
0–100	2	2
100–200	5	7
200–300	x	7 + x
300–400	12	19 + x
400–500	17	36 + x
500–600	20	**56 + x**
600–700	y	56 + x + y
700–800	9	65 + x + y
800–900	7	72 + x + y
900–1000	4	76 + x + y

1. n = 100: 76 + x + y = 100, so **x + y = 24** … (1)
2. Median 525 lies in **500–600**: l = 500, f = 20, cf = 36 + x, h = 100.
3. $525 = 500 + \left(\frac{50 - 36 - x}{20}\right) \times 100 \Rightarrow 25 = (14 - x) \times 5 \Rightarrow 25 = 70 - 5x \Rightarrow$ **x = 9**.
4. From (1): 9 + y = 24 ⇒ **y = 15**.

✍️ Board trick for "find x and y"

One equation always comes from the **total** (Σf = n). The other comes from the **given mean / median / mode**. Two equations, two unknowns. Practise it in the Frequency Detective puzzle.

Mean, median or mode: which one to use?

$\bar{x}$ Mean

Most used. Uses every observation, lies between the smallest and largest, and lets us **compare** two or more distributions (average results of two schools).

Weakness: extreme values pull it. If one class has frequency 2 and the others 20, 25, 20, 21, 18, the mean is a poor representative.

Median

Best when we want a **typical** value and extreme values are present: the **typical productivity** of workers, the **average wage** in a country. A few billionaires do not move it.

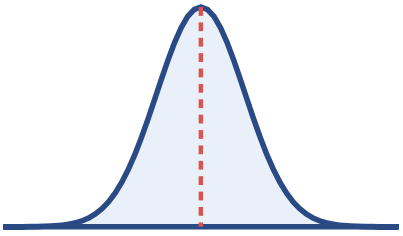
Mode

Best for the **most frequent / most popular** item: the most watched TV programme, the consumer item in greatest demand, the most common colour of vehicles.

3 Median = Mode + 2 Mean

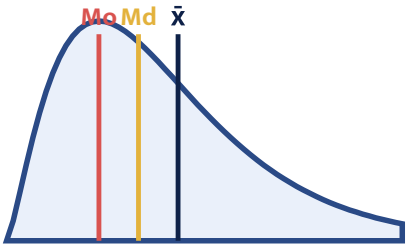
The **empirical relationship** between the three measures (Remark 1). Asked as a 1-mark MCQ almost every year. Rearranged: Mode = 3 Median − 2 Mean · Mean = $\frac{3\text{ Median} - \text{Mode}}{2}$ · Median = $\frac{\text{Mode} + 2\text{ Mean}}{3}$

Symmetric



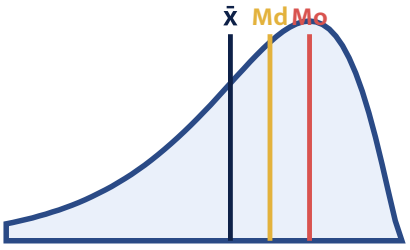
mean = median = mode

Long tail to the right



mode < median < mean

Long tail to the left



mean < median < mode

Shape decides the order. For a symmetric shape all three coincide. When a few very large values stretch the data to the right, they drag the **mean** furthest, the median a little, and the mode not at all. The empirical relation works best for data that is only slightly lopsided.

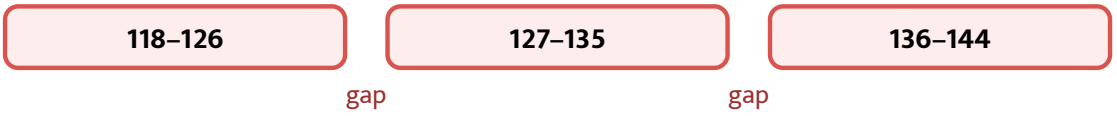
Remark 2 of the book: the median of grouped data with unequal class sizes can also be calculated, but not in Class 10.

Continuous classes (a note to the reader)

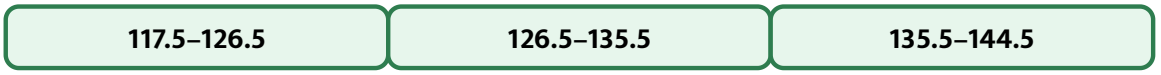
The mode and median formulas assume that the classes are **continuous**: each class starts exactly where the previous one ends. Classes like 118–126, 127–135 leave gaps. Before using the formulas, close the gaps:

1. Gap = lower limit of a class − upper limit of the previous class (127 − 126 = 1).
2. Subtract half the gap from every lower limit and add it to every upper limit.

Given (gaps!)



Continuous (subtract 0.5, add 0.5)



gap = 127 − 126 = 1, so move each limit by half the gap, 0.5

Exercise 13.3, Question 4. The class marks do not change (122, 131, ...), so the **mean** is not affected, but l in the median and mode formulas does change.

Tables

Formula sheet and comparison tables

Measure	Formula (grouped data)	What you need first
Mean, direct	$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$	class marks x_i
Mean, assumed mean	$\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$	$d_i = x_i - a$
Mean, step-deviation	$\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i}$	$u_i = \frac{x_i - a}{h}$
Mode	$l + (\frac{f_1 - f_0}{2f_1 - f_0 - f_2}) \times h$	modal class (highest f)
Median	$l + (\frac{n/2 - cf}{f}) \times h$	cf column, median class
Empirical	3 Median = Mode + 2 Mean	any two measures

Symbol	In the mode formula	In the median formula
l	lower limit of the **modal** class	lower limit of the **median** class
h	class size	class size
f / f_1	f_1 = frequency of modal class	f = frequency of median class

Symbol	In the mode formula	In the median formula
f_0, f_2	frequencies of the classes before and after	not used
cf	not used	**cumulative** frequency of the class **before** the median class
n	not used	Σf, the total frequency

	Mean	Median	Mode
Uses all values?	Yes	No (only position)	No (only the peak)
Affected by extreme values?	**Yes, strongly**	Hardly	No
Best for	comparing groups	typical value, wages	most popular item
Class used	all classes	median class (cf just above n/2)	modal class (largest f)
Book examples	Ex. 1, 2, 3	Ex. 7, 8	Ex. 4, 5, 6



Common mistakes

❌ **Taking the class with the highest cumulative frequency as the median class**

The last class always has the highest cf. The median class is the one whose cf is the **first to go above n/2**.

❌ **Using the median class's own cf**

In the median formula, cf is the cumulative frequency of the class **before** the median class (22 in Table 13.15, not 29).

❌ **Swapping f_0 and f_2**

f_0 is the class **before**, f_2 the class **after**. The numerator is $f_1 - f_0$. Swapping them gives a wrong mode even though the denominator stays the same.

❌ **Using the upper limit for l**

l is always the **lower** limit of the modal or median class.

❌ Using a "less than" table directly

"Less than 30: 27" is a cumulative frequency. Subtract to get class frequencies first (Example 7, CBSE 2023). The modal class in CBSE 2023 Q11 comes from the class frequencies 3, 9, 15, 30, 18, 5, not from the cfs.

❌ Forgetting to make classes continuous

For 118–126, 127–135 ... use 117.5–126.5 ... before finding the mode or median (Exercise 13.3 Q4, Ex. 13.1 Q5).

❌ Forgetting to add a back

$\Sigma f_i u_i \div \Sigma f_i$ is $\bar{u}$, not $\bar{x}$. Multiply by h and **add a**.

✅ Quick check

The mean, median and mode must lie **inside the range of the data**, and the mode must lie **inside the modal class**. If your mode for modal class 40–55 is 58, re-check f_0 and f_2.

CBSE Previous year board questions

Real Statistics questions from the CBSE Class 10 Mathematics (Standard) board papers, set 30/1/1, with full answers. Wording is lightly shortened; data and options are as in the paper. Filter by year.

– **2026 · 30/1/1 · Q17 · 1 mark**

The mean of a data is 12 and its mode is 21. The median of the data is: (a) 6 (b) 13.5 (c) 15 (d) 14

(c) 15. 3 Median = Mode + 2 Mean = 21 + 24 = 45, so Median = 15.

– 2026 · 30/1/1 · Q35 · 5 marks

The mean of the following frequency distribution is 35 and the total frequency is 25. Find the missing frequencies x and y.

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	1	x	5	7	y	3	1

Class marks: 5, 15, 25, 35, 45, 55, 65.

Total: $1 + x + 5 + 7 + y + 3 + 1 = 25 \Rightarrow \mathbf{x + y = 8}$ ... (1)

$\Sigma f_i x_i = 5 + 15x + 125 + 245 + 45y + 165 + 65 = 605 + 15x + 45y$.

Mean: $\frac{605 + 15x + 45y}{25} = 35 \Rightarrow 15x + 45y = 270 \Rightarrow \mathbf{x + 3y = 18}$ ... (2)

(2) − (1): $2y = 10 \Rightarrow \mathbf{y = 5}$, and $\mathbf{x = 3}$.

– 2025 · 30/1/1 · Q12 · 1 mark

If the mode and mean of a data are 15x and 18x respectively, then the median of the data is: (A) x (B) 11x (C) 17x (D) 34x

(C) 17x. 3 Median = 15x + 2(18x) = 51x ⇒ Median = 17x.

– 2025 · 30/1/1 · Q35 · 5 marks

The mean of the following distribution of daily pocket allowance is ₹18. Find the missing frequency f. Also find the mode.

Allowance (₹)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Children	7	6	9	13	f	5	4

Class marks 12, 14, 16, 18, 20, 22, 24. $\Sigma f = 44 + f$, $\Sigma f_i x_i = 752 + 20f$.

$\frac{752 + 20f}{44 + f} = 18 \Rightarrow 752 + 20f = 792 + 18f \Rightarrow \mathbf{f = 20}$. (This is NCERT Exercise 13.1 Q3!)

Maximum frequency = 20, so the **modal class is 19–21**. $l = 19$, $h = 2$, $f_1 = 20$, $f_0 = 13$, $f_2 = 5$.

Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 19 + \frac{20 - 13}{40 - 13 - 5} \times 2 = 19 + \frac{14}{22} = \mathbf{19.64}$

Careful: with f = 20 the modal class is 19–21, not 17–19.

– 2024 · 30/1/1 · Q9 · 1 mark

For the data $x_1, x_2, ..., x_n$ with frequencies $f_1, f_2, ..., f_n$, the value of $\Sigma f_i(x_i - \bar{x})$ is: (A) $n\bar{x}$ (B) $1/n$ (C) Σf_i (D) 0

(D) 0. $\Sigma f_i(x_i - \bar{x}) = \Sigma f_i x_i - \bar{x}\Sigma f_i = \bar{x}\Sigma f_i - \bar{x}\Sigma f_i = 0$. The deviations from the mean always balance out, like the see-saw.

– 2024 · 30/1/1 · Q14 · 1 mark

The middle-most observation of a data arranged in order is called: (A) mode (B) median (C) mean (D) deviation

(B) median.

– 2024 · 30/1/1 · Q37 · case study, 4 marks

In a game of BINGO, 48 numbers were called out and grouped as below. (i) Write the median class. (iii)(a) Find the median. OR (iii)(b) Find the mode. (Part (ii), on probability, is not from this chapter.)

Number called	0–15	15–30	30–45	45–60	60–75
Frequency	8	9	10	12	9

(i) **Median class 30–45** (cf 27 is the first above n/2 = 24).

(iii)(a)

Number	Frequency f	Cumulative frequency cf
0–15	8	8
15–30	9	17
30–45	10	**27**
45–60	12	39
60–75	9	48
Total	**48**	

n = 48, $\frac{n}{2}$ = 24. The first cf greater than 24 is 27, so the **median class is 30–45**. l = 30, cf = 17, f = 10, h = 15.

Median = $l + \frac{n/2 - cf}{f} \times h = 30 + \frac{24 - 17}{10} \times 15 = 30 + 10.5 =$ **40.5**

(iii)(b)

Maximum frequency = 12, so the **modal class is 45–60**. l = 45, h = 15, $f_1 = 12$, $f_0 = 10$, $f_2 = 9$.

Mode = $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 45 + \frac{12 - 10}{24 - 10 - 9} \times 15 = 45 + \frac{30}{5} =$ **51**

– 2023 · 30/1/1 · Q11 · 1 mark

Marks of 80 students: less than 10 → 3, less than 20 → 12, less than 30 → 27, less than 40 → 57, less than 50 → 75, less than 60 → 80. The modal class is: (A) 10–20 (B) 20–30 (C) 30–40 (D) 50–60

(C) 30–40. Class frequencies: 3, 12 − 3 = 9, 27 − 12 = 15, 57 − 27 = **30**, 75 − 57 = 18, 80 − 75 = 5. The largest is 30, in 30–40.

– 2023 · 30/1/1 · Q15 · 1 mark

If every observation of a data is increased by 3, then the mean: (A) remains unchanged (B) increases by 3 (C) increases by 6 (D) increases by 3n

(B) increases by 3. The new sum is $\Sigma x + 3n$, so the new mean = $\bar{x} + 3n/n = \bar{x} + 3$.

– 2023 · 30/1/1 · Q37 · case study, 4 marks

Monsoon rainfall (2018) of 24 sub-divisions of India: (I) Write the modal class. (II) Find the median rainfall. OR Find the mean rainfall. (III) How many sub-divisions had at least 1000 mm of rainfall?

Rainfall (mm)	200–400	400–600	600–800	800–1000	1000–1200	1200–1400	1400–1600	1600
Sub-divisions	2	4	7	4	2	3	1	

(I) Highest frequency 7 → **modal class 600–800**.

(II)

Rainfall (mm)	Frequency f	Cumulative frequency cf
200–400	2	2
400–600	4	6
600–800	7	**13**
800–1000	4	17
1000–1200	2	19
1200–1400	3	22
1400–1600	1	23
1600–1800	1	24
Total	**24**	

n = 24, $\frac{n}{2}$ = 12. The first cf greater than 12 is 13, so the **median class is 600–800**. l = 600, cf = 6, f = 7, h = 200.

Median = $l + \frac{n/2 - cf}{f} \times h = 600 + \frac{12 - 6}{7} \times 200 = 600 + 171.429 =$ **771.43**

OR mean:

Rainfall (mm)	f_i	x_i	$f_i x_i$
200–400	2	300	600
400–600	4	500	2000
600–800	7	700	4900
800–1000	4	900	3600

Rainfall (mm)	f_i	x_i	$f_i x_i$
1000–1200	2	1100	2200
1200–1400	3	1300	3900
1400–1600	1	1500	1500
1600–1800	1	1700	1700
Total	**$\Sigma f_i = 24$**		**$\Sigma f_i x_i = 20400$**

Direct method: $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{20400}{24} =$ **850**

(III) 1000 mm or more: 2 + 3 + 1 + 1 = **7 sub-divisions**.

Practice

Board-pattern practice

MCQs (1 mark)

– **M1** **The class mark of the class 10–25 is (a) 15 (b) 17.5 (c) 12.5 (d) 35**

(b) 17.5 = (10 + 25) ÷ 2.

– **M2** **If the mean of a data is 20 and the median is 22, the mode is (a) 26 (b) 24 (c) 21 (d) 18**

(a) 26: Mode = 3(22) − 2(20) = 66 − 40.

– **M3** **For the distribution 0–5, 5–10, 10–15, 15–20, 20–25 with frequencies 10, 15, 12, 20, 9, the sum of the lower limits of the median class and the modal class is (a) 15 (b) 25 (c) 30 (d) 35**

(b) 25. n = 66, n/2 = 33, cf: 10, 25, 37 → median class 10–15. Modal class 15–20. 10 + 15 = 25.

– **M4** **In the formula $\bar{x} = a + h\,\bar{u}$, u_i is (a) $x_i - a$ (b) $(x_i - a)/h$ (c) $h(x_i - a)$ (d) $(a - x_i)/h$**

(b)

– **M5** **Which measure cannot be found exactly from a histogram alone? (a) mode (b) median (c) mean (d) all three are approximate for grouped data**

(d). All measures of grouped data are estimates, because of the mid-point / even-spread assumption.

Assertion–Reason

(a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

– **AR1** **A: The mean of a grouped data found by the direct, assumed mean and step-deviation methods is the same. R: The assumed mean and step-deviation methods are simplified forms of the direct method.**

(a)

– **AR2** **A: The median class is the class with the maximum frequency. R: The median divides the data into two equal halves.**

(d). The class with the maximum frequency is the **modal** class; the median class has cf just above n/2.

– **AR3** **A: If the mode is 52 and the mean is 62, then the median is 58.67. R: 3 Median = Mode + 2 Mean.**

(a): (52 + 124) ÷ 3 = 58.67.

Short answer (2–3 marks)

– **S1** **Find the mode of the data.**

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	8	15	10	2

Maximum frequency = 15, so the **modal class is 20–30**. l = 20, h = 10, $f_1 = 15$, $f_0 = 8$, $f_2 = 10$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{15 - 8}{30 - 8 - 10} \times 10 = 20 + \frac{70}{12} = \mathbf{25.83}$$

– **S2** **The mean of the following distribution is 50. Find p.**

Class	0–20	20–40	40–60	60–80	80–100
Frequency	17	p	32	24	19

Class marks 10, 30, 50, 70, 90. $\Sigma f = 92 + p$, $\Sigma f_i x_i = 170 + 30p + 1600 + 1680 + 1710 = 5160 + 30p$.

$\frac{5160 + 30p}{92 + p} = 50 \Rightarrow 5160 + 30p = 4600 + 50p \Rightarrow 20p = 560 \Rightarrow \mathbf{p = 28}$.

– **S3** **Convert into a frequency table and name the modal class: less than 20 → 4, less than 40 → 16, less than 60 → 40, less than 80 → 52, less than 100 → 60.**

Frequencies: 0–20: 4, 20–40: 12, 40–60: **24**, 60–80: 12, 80–100: 8. Modal class **40–60**; mode = 40 + (12 ÷ 24) × 20 = **50**.

Case study (4 marks)

C1 **MT Classes asked 80 students how many minutes they take to reach the class.**

Time (min)	0–10	10–20	20–30	30–40	40–50	50–60
Students	4	10	16	24	18	8

(i) Write the modal class and the median class. (ii) Find the median time. (iii) Find the mode. OR Find the mean time.

(i) Modal class **30–40** (24 students). n/2 = 40, cf 4, 14, 30, 54 → median class **30–40** too.

(ii)

Time (min)	Frequency f	Cumulative frequency cf
0–10	4	4
10–20	10	14
20–30	16	30
30–40	24	**54**
40–50	18	72
50–60	8	80
Total	**80**	

n = 80, $\frac{n}{2}$ = 40. The first cf greater than 40 is 54, so the **median class is 30–40**. l = 30, cf = 30, f = 24, h = 10.

Median = $l + \frac{n/2 - cf}{f} \times h = 30 + \frac{40 - 30}{24} \times 10 = 30 + 4.167 =$ **34.17 minutes**

(iii)

Maximum frequency = 24, so the **modal class is 30–40**. l = 30, h = 10, $f_1 = 24$, $f_0 = 16$, $f_2 = 18$.

Mode = $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{24 - 16}{48 - 16 - 18} \times 10 = 30 + \frac{80}{14} =$ **35.71 minutes**

OR

Time (min)	f_i	x_i	$u_i = \frac{x_i - 35}{10}$	f_iu_i
0–10	4	5	−3	−12
10–20	10	15	−2	−20

Time (min)	f_i	x_i	$u_i = \frac{x_i - 35}{10}$	$f_i u_i$
20–30	16	25	−1	−16
30–40	24	35	0	0
40–50	18	45	1	18
50–60	8	55	2	16
Total	**80**			**−14**

Step-deviation method with a = 35, h = 10: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 35 + 10 \times \frac{-14}{80} =$ **33.25**

Recap Summary (NCERT 13.5)

1. The mean of grouped data: (i) direct method $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$ (ii) assumed mean method $\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$ (iii) step-deviation method $\bar{x} = a + (\frac{\Sigma f_i u_i}{\Sigma f_i}) \times h$, assuming the frequency of each class is centred at its mid-point, the **class mark**.
2. The mode of grouped data: Mode $= l + (\frac{f_1 - f_0}{2f_1 - f_0 - f_2}) \times h$.
3. The **cumulative frequency** of a class is the frequency obtained by adding the frequencies of all the classes preceding it (and the class itself).
4. The median of grouped data: Median $= l + (\frac{n/2 - cf}{f}) \times h$.
5. For the mode and median, the classes must be **continuous**. 3 Median = Mode + 2 Mean.

Quiz Test yourself

Tap an answer. **Score: 0 / 20**

1. The class mark of 25–40 is

- 15
- **32.5**
- 65
- 30

(25 + 40) ÷ 2.

2. In the direct method, $\bar{x}$ =

- $\Sigma x_i/n$
- **$\Sigma f_i x_i/\Sigma f_i$**
- $\Sigma f_i/\Sigma x_i$
- $\Sigma f_i d_i$

Sum of f × x over the sum of f.

3. Grouped mean 62 vs exact mean 59.3 differ because of

- a wrong formula
- **the mid-point assumption**
- rounding
- the choice of a

Example 1.

4. In the assumed mean method, d_i =

- $x_i + a$
- $a - x_i$
- **$x_i - a$**
- x_i/a

Deviation from a.

5. Changing the assumed mean a changes the mean

- by a
- by h
- **not at all**
- by a/h

Activity 1.

6. In step-deviation, $\bar{x}$ =

- a + ū
- **a + hū**
- h + aū
- a − hū

Multiply ū by h and add a.

7. a = 47.5, h = 15, $\Sigma f_i u_i$ = 29, Σf_i = 30. $\bar{x}$ =

- **62**
- 61.5
- 48.5
- 76.5

47.5 + 15 × 29/30 = 62.

8. The class with the highest frequency is the

- median class
- **modal class**
- mean class
- last class

9. In the mode formula, f_0 is the frequency of the class

- **before the modal class**
- after the modal class
- modal class
- first class

Preceding class.

10. Modal class 40–55, $f_1 = 7$, $f_0 = 3$, $f_2 = 6$. Mode =

- **52** ✓
- 55
- 47.5
- 62

Example 6.

11. Family sizes 1–3: 7, 3–5: 8, 5–7: 2. Mode ≈

- **3.29** ✓
- 4
- 3.5
- 8

Example 5.

12. The median class is the class whose cf is

- the largest
- **just greater than n/2** ✓
- equal to n
- the smallest

13. In the median formula, cf is

- cf of the median class
- **cf of the class before the median class** ✓
- total frequency
- frequency of the median class

A favourite trap.

14. n = 53, l = 60, cf = 22, f = 7, h = 10. Median ≈

- **66.4**
- 63.2
- 64.5
- 70

60 + 45/7.

15. “Less than” cumulative frequencies use the class

- lower limits
- **upper limits**
- class marks
- widths

Table 13.13.

16. 3 Median =

- **Mode + 2 Mean**
- 2 Mode + Mean
- Mode − 2 Mean
- Mean + Mode

Empirical relation.

17. Mean 12, mode 21. Median =

- 13.5
- **15**
- 14
- 6

CBSE 2026.

18. Best measure for “the most popular shoe size”

mean	median
mode	range

Most frequent item.

19. Classes 118–126, 127–135 become

118–127, 127–136	**117.5–126.5, 126.5–135.5**
118.5–126.5, …	no change needed

Subtract and add 0.5.

20. Each observation is increased by 3. The mean

is unchanged	**increases by 3**
is tripled	increases by 3n

CBSE 2023.

NCERT Answers to all 22 exercise questions

Every question of Exercises 13.1, 13.2 and 13.3 with the full table and working you can write in the exam. Tap to open.

Exercise 13.1 (mean)

1. **Number of plants in 20 houses. Find the mean number of plants per house. Which method did you use, and why?**

Plants	0–2	2–4	4–6	6–8	8–10	10–12	12–14
Houses	1	2	1	5	6	2	3

Plants	f_i	x_i	$f_i x_i$
0–2	1	1	1
2–4	2	3	6
4–6	1	5	5
6–8	5	7	35
8–10	6	9	54
10–12	2	11	22
12–14	3	13	39
Total	**$\Sigma f_i = 20$**		**$\Sigma f_i x_i = 162$**

Direct method: $\bar{X} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{162}{20} = \mathbf{8.1}$

Mean = **8.1 plants**. We used the **direct method** because x_i and f_i are small.

2. **Daily wages of 50 workers. Find the mean daily wages by an appropriate method.**

Wages (₹)	500–520	520–540	540–560	560–580	580–600
Workers	12	14	8	6	10

Wages (₹)	f_i	x_i	$u_i = \frac{x_i - 550}{20}$	f_iu_i
500–520	12	510	−2	−24
520–540	14	530	−1	−14
540–560	8	550	0	0
560–580	6	570	1	6
580–600	10	590	2	20
Total	**50**			**−12**

Step-deviation method with a = 550, h = 20: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 550 + 20 \times \frac{-12}{50} =$ **545.2**

Mean daily wages = **₹545.20** (step-deviation, because the x_i are large).

3. **The mean pocket allowance is ₹18. Find the missing frequency f.**

Allowance (₹)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Children	7	6	9	13	f	5	4

Class	f_i	x_i	f_ix_i
11–13	7	12	84
13–15	6	14	84
15–17	9	16	144
17–19	13	18	234
19–21	f	20	20f
21–23	5	22	110
23–25	4	24	96
Total	**44 + f**		**752 + 20f**

$\frac{752 + 20f}{44 + f} = 18 \Rightarrow 752 + 20f = 792 + 18f \Rightarrow 2f = 40 \Rightarrow \mathbf{f = 20}$.

4. **Heartbeats per minute of 30 women. Find the mean.**

Heartbeats/min	65–68	68–71	71–74	74–77	77–80	80–83	83–86
Women	2	4	3	8	7	4	2

Heartbeats	f_i	x_i	$u_i = \frac{x_i - 75.5}{3}$	$f_i u_i$
65–68	2	66.5	−3	−6
68–71	4	69.5	−2	−8
71–74	3	72.5	−1	−3
74–77	8	75.5	0	0
77–80	7	78.5	1	7
80–83	4	81.5	2	8
83–86	2	84.5	3	6
Total	**30**			**4**

Step-deviation method with a = 75.5, h = 3: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 75.5 + 3 \times \frac{4}{30} =$ **75.9**

Mean = **75.9 beats per minute**.

5. **Mangoes in packing boxes. Find the mean number of mangoes in a box. Which method did you choose?**

Mangoes	50–52	53–55	56–58	59–61	62–64
Boxes	15	110	135	115	25

Class marks: 51, 54, 57, 60, 63 (they do not change if the classes are made continuous, 49.5–52.5 …).

Mangoes	f_i	x_i	$u_i = \frac{x_i - 57}{3}$	$f_i u_i$
50–52	15	51	−2	−30
53–55	110	54	−1	−110
56–58	135	57	0	0
59–61	115	60	1	115
62–64	25	63	2	50
Total	**400**			**25**

Step-deviation method with a = 57, h = 3: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 57 + 3 \times \frac{25}{400} =$ **57.19**

Mean = **57.19 mangoes**. Step-deviation, because the frequencies are large and the d_i are multiples of 3.

6. **Daily expenditure on food of 25 households. Find the mean.**

Expenditure (₹)	100–150	150–200	200–250	250–300	300–350
Households	4	5	12	2	2

Expenditure (₹)	f_i	x_i	$u_i = \frac{x_i - 225}{50}$	f_iu_i
100–150	4	125	−2	−8
150–200	5	175	−1	−5
200–250	12	225	0	0
250–300	2	275	1	2
300–350	2	325	2	4
Total	**25**			**−7**

Step-deviation method with a = 225, h = 50: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 225 + 50 \times \frac{-7}{25} =$ **211**

Mean = **₹211**.

7. **Concentration of SO_2 in the air in 30 localities. Find the mean concentration.**

SO_2 (ppm)	0–0.04	0.04–0.08	0.08–0.12	0.12–0.16	0.16–0.2	0.2–0.24
Frequency	4	9	9	2	4	2

SO_2 (ppm)	f_i	x_i	$u_i = \frac{x_i - 0.14}{0.04}$	f_iu_i
0–0.04	4	0.02	−3	−12
0.04–0.08	9	0.06	−2	−18
0.08–0.12	9	0.1	−1	−9
0.12–0.16	2	0.14	0	0
0.16–0.2	4	0.18	1	4
0.2–0.24	2	0.22	2	4
Total	**30**			**−31**

Step-deviation method with a = 0.14, h = 0.04: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 0.14 + 0.04 \times \frac{-31}{30} =$ **0.099**

Mean ≈ **0.099 ppm**.

8. **Absentee record of 40 students. Find the mean number of days a student was absent.**

Days	0–6	6–10	10–14	14–20	20–28	28–38	38–40
Students	11	10	7	4	4	3	1

Days	f_i	x_i	f_ix_i
0–6	11	3	33
6–10	10	8	80
10–14	7	12	84
14–20	4	17	68
20–28	4	24	96
28–38	3	33	99
38–40	1	39	39
Total	**$\Sigma f_i = 40$**		**$\Sigma f_ix_i = 499$**

Direct method: $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{499}{40} =$ **12.47**

Mean = **12.48 days** (classes are unequal, so the direct method is simplest).

9. **Literacy rate of 35 cities. Find the mean literacy rate.**

Literacy rate (%)	45–55	55–65	65–75	75–85	85–95
Cities	3	10	11	8	3

Literacy (%)	f_i	x_i	$u_i = \frac{x_i - 70}{10}$	f_iu_i
45–55	3	50	−2	−6
55–65	10	60	−1	−10
65–75	11	70	0	0
75–85	8	80	1	8
85–95	3	90	2	6
Total	**35**			**−2**

Step-deviation method with a = 70, h = 10: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 70 + 10 \times \frac{-2}{35} =$ **69.43**

Mean literacy rate = **69.43%**.

Exercise 13.2 (mode)

1. **Ages of patients admitted in a hospital. Find the mode and the mean. Compare and interpret.**

Age (years)	5–15	15–25	25–35	35–45	45–55	55–65
Patients	6	11	21	23	14	5

Maximum frequency = 23, so the **modal class is 35–45**. $l = 35$, $h = 10$, $f_1 = 23$, $f_0 = 21$, $f_2 = 14$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 35 + \frac{23 - 21}{46 - 21 - 14} \times 10 = 35 + \frac{20}{11} = \mathbf{36.82 \text{ years}}$$

Age	f_i	x_i	$u_i = \frac{x_i - 30}{10}$	$f_i u_i$
5–15	6	10	−2	−12
15–25	11	20	−1	−11
25–35	21	30	0	0
35–45	23	40	1	23
45–55	14	50	2	28
55–65	5	60	3	15
Total	**80**			**43**

Step-deviation method with $a = 30$, $h = 10$: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 30 + 10 \times \frac{43}{80} = \mathbf{35.38}$

Mode ≈ **36.8 years**, mean ≈ **35.37 years**. The largest number of patients are about 36.8 years old, while the average age of a patient is about 35.37 years.

2. **Lifetimes (in hours) of 225 electrical components. Find the modal lifetime.**

Lifetime (h)	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	10	35	52	61	38	29

Maximum frequency = 61, so the **modal class is 60–80**. $l = 60$, $h = 20$, $f_1 = 61$, $f_0 = 52$, $f_2 = 38$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{61 - 52}{122 - 52 - 38} \times 20 = 60 + \frac{180}{32} = \mathbf{65.625 \text{ hours}}$$

3. Monthly household expenditure of 200 families. Find the modal and the mean monthly expenditure.

Expenditure (₹)	1000–1500	1500–2000	2000–2500	2500–3000	3000–3500	3500–4000
Families	24	40	33	28	30	22

Maximum frequency = 40, so the **modal class is 1500–2000**. $l = 1500$, $h = 500$, $f_1 = 40$, $f_0 = 24$, $f_2 = 33$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 1500 + \frac{40 - 24}{80 - 24 - 33} \times 500 = 1500 + \frac{8000}{23} = \mathbf{1847.83}$$

Expenditure (₹)	f_i	x_i	$u_i = \frac{x_i - 2750}{500}$	f_iu_i
1000–1500	24	1250	−3	−72
1500–2000	40	1750	−2	−80
2000–2500	33	2250	−1	−33
2500–3000	28	2750	0	0
3000–3500	30	3250	1	30
3500–4000	22	3750	2	44
4000–4500	16	4250	3	48
4500–5000	7	4750	4	28
Total	**200**			**−35**

Step-deviation method with $a = 2750$, $h = 500$: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 2750 + 500 \times \frac{-35}{200} = \mathbf{2662.5}$

Modal expenditure ≈ **₹1847.83**, mean expenditure = **₹2662.50**.

4. **State-wise teacher–student ratio. Find the mode and mean, and interpret.**

Students per teacher	15–20	20–25	25–30	30–35	35–40	40–45	45–50	50–55
States/U.T.	3	8	9	10	3	0	0	2

Maximum frequency = 10, so the **modal class is 30–35**. $l = 30$, $h = 5$, $f_1 = 10$, $f_0 = 9$, $f_2 = 3$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{10 - 9}{20 - 9 - 3} \times 5 = 30 + \frac{5}{8} = \mathbf{30.62}$$

Students per teacher	f_i	x_i	$u_i = \frac{x_i - 32.5}{5}$	f_iu_i
15–20	3	17.5	−3	−9
20–25	8	22.5	−2	−16
25–30	9	27.5	−1	−9
30–35	10	32.5	0	0
35–40	3	37.5	1	3
40–45	0	42.5	2	0
45–50	0	47.5	3	0
50–55	2	52.5	4	8
Total	**35**			**−23**

Step-deviation method with $a = 32.5$, $h = 5$: $\bar{X} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 32.5 + 5 \times \frac{-23}{35} = \mathbf{29.21}$

Mode ≈ **30.6**, mean ≈ **29.2**. Most states have about 30.6 students per teacher; on average there are about 29.2 students per teacher.

5. **Runs scored by top batsmen in ODIs. Find the mode.**

Runs	3000–4000	4000–5000	5000–6000	6000–7000	7000–8000	8000–9000	90
Batsmen	4	18	9	7	6	3	

Maximum frequency = 18, so the **modal class is 4000–5000**. $l = 4000$, $h = 1000$, $f_1 = 18$, $f_0 = 4$, $f_2 = 9$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 4000 + \frac{18 - 4}{36 - 4 - 9} \times 1000 = 4000 + \frac{14000}{23} = \mathbf{4608.7 \text{ runs}}$$

6. **Number of cars passing a spot in 100 periods of 3 minutes. Find the mode.**

Cars	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	7	14	13	12	20	11	15	8

Maximum frequency = 20, so the **modal class is 40–50**. $l = 40$, $h = 10$, $f_1 = 20$, $f_0 = 12$, $f_2 = 11$.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \frac{20 - 12}{40 - 12 - 11} \times 10 = 40 + \frac{80}{17} = \mathbf{44.7\ cars}$$

Exercise 13.3 (median)

1. **Monthly electricity consumption of 68 consumers. Find the median, mean and mode and compare them.**

Units	65–85	85–105	105–125	125–145	145–165	165–185	185–205
Consumers	4	5	13	20	14	8	4

Units	Frequency f	Cumulative frequency cf
65–85	4	4
85–105	5	9
105–125	13	22
125–145	20	**42**
145–165	14	56
165–185	8	64
185–205	4	68
Total	**68**	

n = 68, $\frac{n}{2}$ = 34. The first cf greater than 34 is 42, so the **median class is 125–145**. l = 125, cf = 22, f = 20, h = 20.

Median = $l + \frac{n/2 - cf}{f} \times h = 125 + \frac{34 - 22}{20} \times 20 = 125 + 12 =$ **137**

Units	f_i	x_i	$u_i = \frac{x_i - 135}{20}$	$f_i u_i$
65–85	4	75	−3	−12
85–105	5	95	−2	−10
105–125	13	115	−1	−13
125–145	20	135	0	0
145–165	14	155	1	14
165–185	8	175	2	16
185–205	4	195	3	12
Total	**68**			**7**

Step-deviation method with a = 135, h = 20: $\bar{x} = a + h \times \frac{\Sigma f_i u_i}{\Sigma f_i} = 135 + 20 \times \frac{7}{68} =$ **137.06**

Maximum frequency = 20, so the **modal class is 125–145**. l = 125, h = 20, $f_1 = 20$, $f_0 = 13$, $f_2 = 14$.

Mode = $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 125 + \frac{20 - 13}{40 - 13 - 14} \times 20 = 125 + \frac{140}{13} =$ **135.77**

Median = **137 units**, mean ≈ **137.05 units**, mode ≈ **135.76 units**. All three are almost equal, so the data is nearly symmetric.

2. The median of the distribution is 28.5. Find x and y.

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	5	x	20	15	y	5

(total 60)

Class	f	cf
0–10	5	5
10–20	x	5 + x
20–30	**20**	25 + x
30–40	15	40 + x
40–50	y	40 + x + y
50–60	5	45 + x + y

45 + x + y = 60 ⇒ **x + y = 15**. Median 28.5 lies in 20–30: l = 20, cf = 5 + x, f = 20, h = 10, n/2 = 30.

$28.5 = 20 + \frac{30 - 5 - x}{20} \times 10 \Rightarrow 8.5 = \frac{25 - x}{2} \Rightarrow 17 = 25 - x \Rightarrow$ **x = 8, y = 7**.

3. Ages of 100 policy holders (policies only from 18 years to below 60): below 20 → 2, below 25 → 6, below 30 → 24, below 35 → 45, below 40 → 78, below 45 → 89, below 50 → 92, below 55 → 98, below 60 → 100. Find the median age.

Subtract to get class frequencies (first class 18–20).

Age (years)	Frequency f	Cumulative frequency cf
18–20	2	2
20–25	4	6
25–30	18	24
30–35	21	45
35–40	33	**78**
40–45	11	89
45–50	3	92
50–55	6	98
55–60	2	100
Total	**100**	

n = 100, $\frac{n}{2}$ = 50. The first cf greater than 50 is 78, so the **median class is 35–40**. l = 35, cf = 45, f = 33, h = 5.

Median = $l + \frac{n/2 - cf}{f} \times h = 35 + \frac{50 - 45}{33} \times 5 = 35 + 0.758 =$ **35.76 years**

4. **Lengths of 40 leaves: 118–126: 3, 127–135: 5, 136–144: 9, 145–153: 12, 154–162: 5, 163–171: 4, 172–180: 2. Find the median length.**

Make the classes continuous: 117.5–126.5, 126.5–135.5, ...

Length (mm)	Frequency f	Cumulative frequency cf
117.5–126.5	3	3
126.5–135.5	5	8
135.5–144.5	9	17
144.5–153.5	12	**29**
153.5–162.5	5	34
162.5–171.5	4	38
171.5–180.5	2	40
Total	**40**	

n = 40, $\frac{n}{2}$ = 20. The first cf greater than 20 is 29, so the **median class is 144.5–153.5**. l = 144.5, cf = 17, f = 12, h = 9.

Median = $l + \frac{n/2 - cf}{f} \times h = 144.5 + \frac{20 - 17}{12} \times 9 = 144.5 + 2.25 =$ **146.75 mm**

5. **Life time of 400 neon lamps. Find the median life time.**

Life time (h)	1500–2000	2000–2500	2500–3000	3000–3500	3500–4000	4000–4500	4
Lamps	14	56	60	86	74	62	

Life time (h)	Frequency f	Cumulative frequency cf
1500–2000	14	14
2000–2500	56	70
2500–3000	60	130
3000–3500	86	**216**
3500–4000	74	290
4000–4500	62	352
4500–5000	48	400
Total	**400**	

n = 400, $\frac{n}{2}$ = 200. The first cf greater than 200 is 216, so the **median class is 3000–3500**. l = 3000, cf = 130, f = 86, h = 500.

Median = $l + \frac{n/2 - cf}{f} \times h = 3000 + \frac{200 - 130}{86} \times 500 = 3000 + 406.977 =$ **3406.98 hours**

6. Number of letters in 100 surnames. Find the median, the mean and the modal size.

Letters	1–4	4–7	7–10	10–13	13–16	16–19
Surnames	6	30	40	16	4	4

Letters	Frequency f	Cumulative frequency cf
1–4	6	6
4–7	30	36
7–10	40	**76**
10–13	16	92
13–16	4	96
16–19	4	100
Total	**100**	

n = 100, $\frac{n}{2}$ = 50. The first cf greater than 50 is 76, so the **median class is 7–10**. l = 7, cf = 36, f = 40, h = 3.

Median = $l + \frac{n/2 - cf}{f} \times h = 7 + \frac{50 - 36}{40} \times 3 = 7 + 1.05 =$ **8.05**

Letters	f_i	x_i	$f_i x_i$
1–4	6	2.5	15
4–7	30	5.5	165
7–10	40	8.5	340
10–13	16	11.5	184
13–16	4	14.5	58
16–19	4	17.5	70
Total	**$\Sigma f_i = 100$**		**$\Sigma f_i x_i = 832$**

Direct method: $\bar{X} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{832}{100} =$ **8.32**

Maximum frequency = 40, so the **modal class is 7–10**. l = 7, h = 3, $f_1 = 40$, $f_0 = 30$, $f_2 = 16$.

Mode = $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 7 + \frac{40 - 30}{80 - 30 - 16} \times 3 = 7 + \frac{30}{34} =$ **7.88**

Median = **8.05**, mean = **8.32**, mode ≈ **7.88** letters.

7. Weights of 30 students. Find the median weight.

Weight (kg)	40–45	45–50	50–55	55–60	60–65	65–70	70–75
Students	2	3	8	6	6	3	2

Weight (kg)	Frequency f	Cumulative frequency cf
40–45	2	2
45–50	3	5
50–55	8	13
55–60	6	**19**
60–65	6	25
65–70	3	28
70–75	2	30
Total	**30**	

n = 30, $\frac{n}{2}$ = 15. The first cf greater than 15 is 19, so the **median class is 55–60**. l = 55, cf = 13, f = 6, h = 5.

Median = $l + \frac{n/2 - cf}{f} \times h = 55 + \frac{15 - 13}{6} \times 5 = 55 + 1.667 =$ **56.67 kg**

Tables and examples from the NCERT Mathematics textbook for Class 10 (reprint 2026-27). Board questions from CBSE Class 10 Mathematics (Standard) papers 2023 to 2026, set 30/1/1.